

Safety Data Sheet

Melflux® 5581 F MB

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Version: 1.0

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(30857638/SDS_GEN_MX/EN)

1. Identification

Product identifier used on the label

Melflux® 5581 F MB

Recommended use of the chemical and restriction on use

Recommended use*: Product for construction chemicals

Recommended use*: Raw material

Unsuitable for use: None known

Suitable for use in industrial sector: cement industry

* The "Recommended use" identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:

BASF Mexicana S.A. de C.V.

Av. Insurgentes Sur 975

Col. CD. De Los Deportes,

C.P. 03710 Ciudad de México

MÉXICO

Telephone: +52 55 5325 2600

Emergency telephone number

24 Hour Emergency Response Information

SETIQ: 1800-00-214-(Rep. Mexicana) or 55-59-15-88 (CDMX)

Telephone: +1-800-849-5204 or +1-833-229-1000

Other means of identification

Chemical family: Polymer based on: polycarboxylate ether

2. Hazards Identification

According to Regulation NOM-018-STPS-2015

Classification of the product

Combustible Dust

Combustible Dust (1)

Combustible Dust

Label elements

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Signal Word:
Warning

Hazard Statement:
May form combustible dust concentration in air.

Hazards not otherwise classified

The product is under certain conditions capable of dust explosion.

3. Composition / Information on Ingredients

According to Regulation NOM-018-STPS-2015

Silicon dioxide
CAS Number: 7631-86-9
Content (W/W): ≥ 5.0 - $\leq 10.0\%$
Synonym: Siliciumdioxid

The actual concentration is withheld as a trade secret.

4. First-Aid Measures

Description of first aid measures

General advice:
Remove contaminated clothing.

If inhaled:
If difficulties occur after dust has been inhaled, remove to fresh air and seek medical attention.

If on skin:
Wash thoroughly with soap and water. If irritation develops, seek medical attention.

If in eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open. If irritation develops, seek medical attention.

If swallowed:
Rinse mouth and then drink 200-300 ml of water. Do not induce vomiting. Seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: (Further) symptoms and / or effects are not known so far
Hazards: No applicable information available.

Indication of any immediate medical attention and special treatment needed

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Note to physician

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
dry powder, foam

Unsuitable extinguishing media for safety reasons:
carbon dioxide

Additional information:
Avoid whirling up the material/product because of the danger of dust explosion.

Special hazards arising from the substance or mixture

Hazards during fire-fighting:
harmful vapours
Generation of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Advice for fire-fighters

Protective equipment for fire-fighting:
Firefighters should be equipped with self-contained breathing apparatus and turn-out gear.

Further information:

Dusty conditions may ignite explosively in the presence of an ignition source causing flash fire.

6. Accidental release measures

Further accidental release measures:

Avoid dispersal of dust in the air (e.g. by clearing dusty surfaces with compressed air). Avoid the formation and build-up of dust - danger of dust explosion. Dust in sufficient concentration can result in an explosive mixture in air. Handle to minimize dusting and eliminate open flame and other sources of ignition.

Personal precautions, protective equipment and emergency procedures

Avoid dust formation. Use personal protective clothing.

Environmental precautions

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up

For small amounts: Pick up with suitable appliance and dispose of.
For large amounts: Contain with dust binding material and dispose of.
Avoid raising dust.

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7. Handling and Storage

Precautions for safe handling

Closed containers should only be opened in well-ventilated areas.

Protection against fire and explosion:

Dust can form an explosive mixture with air.

Conditions for safe storage, including any incompatibilities

No applicable information available.

Further information on storage conditions: Keep only in the original container in a cool, dry, well-ventilated place away from ignition sources, heat or flame. Protect from direct sunlight. The substance/product may cake at higher temperatures/pressure.

Storage class according to TRGS 510 (originally VCI, Germany): (11) Combustible solids.

Protect from temperatures above: 40 °C

The packed product must be protected against exceeding the indicated temperature.

8. Exposure Controls/Personal Protection

Components with occupational exposure limits

Silicon dioxide	OEL, MX:	TWA value	3 mg/m ³	Respirable fraction ;
	OEL, MX:	TWA value	10 mg/m ³	Inhalable fraction ;

Advice on system design:

It is recommended that all dust control equipment such as local exhaust ventilation and material transport systems involved in handling of this product contain explosion relief vents or an explosion suppression system or an oxygen deficient environment. Ensure that dust-handling systems (such as exhaust ducts, dust collectors, vessels, and processing equipment) are designed in a manner to prevent the escape of dust into the work area (i.e., there is no leakage from the equipment). Use only appropriately classified electrical equipment and powered industrial trucks.

Personal protective equipment

Respiratory protection:

Wear a NIOSH-certified (or equivalent) particulate respirator.

Hand protection:

Chemical resistant protective gloves

Eye protection:

Safety glasses with side-shields. Wear face shield if splashing hazard exists.

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. head protection, apron, protective boots, chemical-protection suit.

General safety and hygiene measures:

Wear protective clothing as necessary to minimize contact. Handle in accordance with good industrial hygiene and safety practice. When using, do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift. At the end of the shift the skin should be

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cleaned and skin-care agents applied. Gloves must be inspected regularly and prior to each use.
Replace if necessary (e.g. pinhole leaks).

9. Physical and Chemical Properties

Physical state:	solid	
Form:	powder	
Odour:	No data available.	
Odour threshold:	No data available.	
Colour:	Yellowish to brownish	
pH value:	approx. 6.5 - 8.5 (20 %(m), 20 °C)	
Melting temperature:	The substance / product decomposes therefore not determined.	
Freezing point:	No data available.	
boiling temperature:	not applicable	
Sublimation point:	No data available.	
Flash point:	not applicable	
Autoignition:	360 °C	(BS EN 50281-2-1 Method B)
Vapour pressure:	The product has not been tested.	
Density:	not applicable	
Relative density:	No data available.	
Bulk density:	approx. 300 - 550 kg/m ³	
Relative vapour density:	The product is a non-volatile solid.	
Partitioning coefficient n-octanol/water (log Pow):	not applicable for mixtures	
Self-ignition temperature:	165 °C Data for powdery solid. No self ignition was observed up to the specified temperature.	(VDI 2263, sheet 1, 1.4.2 (May 1990))
Thermal decomposition:	> 180 °C No decomposition if stored and handled as prescribed/indicated.	
Viscosity, dynamic:	not applicable, the product is a solid	
Viscosity, kinematic:	No applicable information available.	
Solubility in water:	(20 °C) soluble	
Miscibility with water:	not applicable	
Solubility (qualitative):	No data available.	
Molecular weight:	No data available.	
Evaporation rate:	The product is a non-volatile solid.	
Other Information:	If necessary, information on other physical and chemical parameters is indicated in this section.	
Particle characteristics		
Particle size distribution:	< 300 µm	(D50, Volumetric Distribution, ISO 13322-2)

10. Stability and Reactivity

Reactivity

No hazardous reactions if stored and handled as prescribed/indicated.

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Corrosion to metals:
Corrosive effects to metal are not anticipated.

Oxidizing properties:
not fire-propagating

Minimum ignition energy:
30 - 100 mJ, Inductivity: 1 mH, Grain size distribution: < 63 µm

Chemical stability

The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions

Dust explosion hazard.

Conditions to avoid

See SDS section 7 - Handling and storage.

Incompatible materials

strong acids, strong bases, strong oxidizing agents, strong reducing agents

Hazardous decomposition products

Decomposition products:
Hazardous decomposition products: No hazardous decomposition products if stored and handled as prescribed/indicated.

Thermal decomposition:
> 180 °C
No decomposition if stored and handled as prescribed/indicated.

11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Acute toxicity

Assessment of acute toxicity: Virtually nontoxic after a single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact. Based on available data, the classification criteria are not met.

Oral

Type of value: LD50

Species: rat

Value: > 5,000 mg/kg

The product has not been tested. The statement has been derived from the properties of the individual components.

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Inhalation

Type of value: LC50
Species: rat
Exposure time: 4 h
not determined

Dermal

Type of value: LD50
Species: rat
not determined

Assessment other acute effects

Assessment of STOT single:

Based on available data, the classification criteria are not met.

Irritation / corrosion

Assessment of irritating effects: No irritation is expected under intended use and appropriate handling. Based on available data, the classification criteria are not met.

Skin

Species: rabbit

Result: non-irritant

The product has not been tested. The statement has been derived from the properties of the individual components.

Eye

Species: rabbit

Result: non-irritant

The product has not been tested. The statement has been derived from the properties of the individual components.

Sensitization

Assessment of sensitization: Based on the ingredients, there is no suspicion of a skin-sensitizing potential. Based on available data, the classification criteria are not met.

Species: guinea pig

Result: Non-sensitizing.

Aspiration Hazard

not applicable

Chronic Toxicity/Effects

Repeated dose toxicity

Assessment of repeated dose toxicity: The information available on the product provides no indication of toxicity on target organs after repeated exposure.

Genetic toxicity

Assessment of mutagenicity: Based on the ingredients, there is no suspicion of a mutagenic effect.

Carcinogenicity

Assessment of carcinogenicity: The whole of the information assessable provides no indication of a carcinogenic effect.

Reproductive toxicity

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Assessment of reproduction toxicity: Based on the ingredients, there is no suspicion of a toxic effect on reproduction.

Teratogenicity

Assessment of teratogenicity: Based on the ingredients, there is no suspicion of a teratogenic effect.

12. Ecological Information

Toxicity

Aquatic toxicity

Assessment of aquatic toxicity:

Based on available data, the classification criteria are not met. There is a high probability that the product is not acutely harmful to aquatic organisms.

Toxicity to fish

LC50 (96 h), Fish (other)
not determined

Aquatic invertebrates

LC50 (48 h), daphnia (other)
not determined

Aquatic plants

EC50 (72 h), algae (other)
not determined

Chronic toxicity to fish

No data available.

Chronic toxicity to aquatic invertebrates

No data available.

Microorganisms/Effect on activated sludge

Toxicity to microorganisms

other
bacteria/EC50 (0.5 h):
not determined

Persistence and degradability

Assessment biodegradation and elimination (H₂O)

Not readily biodegradable (by OECD criteria).

Bioaccumulative potential

Assessment bioaccumulation potential

No data available.

Mobility in soil

Assessment transport between environmental compartments

No data available.

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Additional information

Other ecotoxicological advice:

Do not discharge product into the environment without control. The product has not been tested. The statements on ecotoxicology have been derived from the properties of the individual components.

13. Disposal considerations

Waste disposal of substance:

Dispose of in accordance with local authority regulations. Do not discharge into drains/surface waters/groundwater.

Container disposal:

Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.

14. Transport Information

Land transport

TDG

Not classified as a dangerous good under transport regulations

Sea transport

IMDG

Not classified as a dangerous good under transport regulations

Air transport

IATA/ICAO

Not classified as a dangerous good under transport regulations

15. Regulatory Information

Federal Regulations

Not applicable

NFPA Hazard codes:

Health: 1 Fire: 1 Reactivity: 0 Special:

16. Other Information

SDS Prepared by:

BASF NA Product Regulations
SDS Prepared on: 2026/02/17

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This information is considered accurate but is not exhaustive and shall only be used as a guideline based on current knowledge of the chemical substance or mixture. Safety precautions suitable for the product must be applied.

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Date / Previous version: not applicable

Previous version: none

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